Population-level impact of vaginal microbiome restoration on HIV acquisition and preterm birth in South Africa: a modelling study

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ABSTRACT

Background Bacterial vaginosis (BV) is highly prevalent in sub-Saharan Africa and increases risk of both HIV acquisition and preterm birth (PTB). Standard antibiotic treatment provides only transient benefit with high recurrence rates. Live biotherapeutic products (LBPs) designed to establish protective *Lactobacillus crispatus*-dominant vaginal microbiota represent a promising intervention, but their population-level impact remains uncertain. A fundamental challenge in evaluating LBPs is the disconnect between what clinical trials can feasibly measure and what drives long-term population health outcomes — a gap that mathematical modeling is uniquely positioned to bridge.

Methods We developed a stochastic agent-based model simulating vaginal microbiome dynamics, HIV transmission, and pregnancy outcomes in South Africa from 1990 to 2050. We evaluated hypothetical LBP deployment beginning in 2035 across product characteristics including efficacy of establishing CST I (50–80%), duration of CST I protection (6–18 months), and fitness of LBP strain, with delivery through existing syndromic and prenatal care pathways. We used model outputs to translate trial-observable endpoints into projected population health impact and to identify quantitative target product profile (TPP) thresholds. Primary outcomes were cumulative HIV infections and preterm births averted between 2035 and 2050.

Findings Under plausible product assumptions (80% efficacy, 12-month protection duration), LBP introduction in 2035 was projected to avert 173,000 HIV infections (6.5% reduction, 95% UI: 127,000–209,000) and 251,000 preterm births (11.1% reduction, 95% UI: 241,000–274,000) by 2050. Across the full range of product characteristics, impact varied from 42,000 to 249,000 HIV infections averted (1.5–9.0% reduction) and 177,000 to 258,000 preterm births averted (7.6–11.2% reduction). Meaningful HIV prevention required protection lasting ≥12 months, whereas PTB reduction was achievable with shorter-acting products. Duration of protection was the dominant driver of HIV impact; efficacy was the dominant driver of PTB impact. Nugent score was identified as an imperfect proxy for *L. crispatus* (CST I) establishment — the mechanistic driver of population health impact — with divergence increasing for products with lower CST I prevalence.

Interpretation LBPs targeting vaginal dysbiosis could meaningfully reduce both HIV and preterm birth in high-burden settings. Product development should prioritize durable colonization lasting ≥12 months for HIV prevention impact; PTB prevention is achievable across a broader range of product profiles. Phase III trials should measure long-term colonization

persistence, CST classification by metagenomics, and efficacy in women with refractory dysbiosis to generate data translatable into population impact projections and to validate these exploratory TPP targets.

Funding: Gates Foundation

Keywords: vaginal microbiome, bacterial vaginosis, live biotherapeutic products, HIV prevention, preterm birth, agent-based modelling, target product profile, South Africa

INTRODUCTION

Bacterial vaginosis (BV) affects 20–50% of women in sub-Saharan Africa and is characterized by a dysbiotic shift from protective *Lactobacillus crispatus*-dominant vaginal microbiota to diverse anaerobic bacteria (1–4). This dysbiosis increases HIV acquisition risk 1.5–3-fold through epithelial disruption, antimicrobial peptide depletion, and genital inflammation (7–10), and increases preterm birth risk 1.5–2-fold through ascending infection and inflammatory pathways (11–14). In South Africa, where HIV prevalence among reproductive-age women exceeds 20%, preterm birth rates approach 15%, and BV prevalence reaches 60% (1,2,13,15–17,34,40), this dual burden creates substantial population-level impact.

Standard metronidazole or clindamycin treatment offers temporary relief but has recurrence rates exceeding 50% within 6–12 months (5,6,18), largely due to failure to restore protective *Lactobacillus* species. Live biotherapeutic products (LBPs) represent a paradigm shift from anaerobic suppression to active restoration of *L. crispatus*-dominant microbiota. As regulated biologics under FDA oversight (19,20), LBPs are designed for strain-specific engraftment and sustained protection through lactic acid production, competitive exclusion, and pH maintenance (3,21,22). Early trials show promise for BV recurrence prevention (23,24), with several products in late-stage development.

A fundamental challenge in evaluating LBPs is the disconnect between what clinical trials can feasibly measure and what drives population-level health outcomes. Trials measure BV recurrence at 3–6 months and microbiome dynamics by Nugent score or metagenomics (3,25,26) — endpoints observable within feasible timescales. However, HIV prevention impact accumulates over years of sexual exposure, making it infeasible to power trials for incidence endpoints, and PTB prevention requires large trials to detect significant effects (13,14). This temporal and scale mismatch creates two strategic risks: trial "success" may misrepresent population value if products fail to sustain *L. crispatus* dominance beyond 12 months; and products with modest short-term signals but strong long-term stability (25,26) may be deprioritized based on early readouts alone. Mathematical modeling bridges this gap — translating observable trial endpoints into long-term population health projections, and working in reverse to derive quantitative TPP targets and trial design recommendations. Prior modeling has not captured the dynamic, recurrent nature of vaginal dysbiosis (25,26,36,37) or its interactions with HIV and pregnancy across realistic delivery pathways.

We developed an agent-based model to: (1) estimate population-level LBP impact under varying product characteristics and delivery strategies; (2) identify the critical product attributes and

derive quantitative TPP thresholds; (3) assess the relationship between trial-observable endpoints and CST I prevalence; and (4) determine how delivery strategy modifies population-level reach.

METHODS

Study design and population We developed a stochastic agent-based model simulating 100,000 agents in South Africa from 1990 to 2050, comparing standard-of-care metronidazole to LBP introduction in 2035. The model was implemented using Starsim (43). Primary outcomes were cumulative HIV infections and preterm births averted from 2035–2050.

Model structure Female agents were assigned an underlying microbiome predisposition at sexual debut: 15% optimal (favoring CST I) and 85% non-optimal, reflecting longitudinal cohort distributions (3,25,26,27,34). The microbiome was represented using three CSTs: CST I (*L. crispatus*-dominant), CST III (*L. iners*-dominant), and CST IV (diverse anaerobes, BV). Women experienced stochastic monthly transitions toward dysbiotic states, calibrated to reproduce 60% BV prevalence and 50% recurrence following metronidazole (1,2,5,6), modified by sexual activity and partner circumcision status. HIV transmission occurred through sexual and vertical routes, calibrated to South African prevalence trends 1990–2020 (38,39), with CST-specific transmission multipliers based on meta-analyses (7,8): CST III (1.5× acquisition, 1.75× transmission) and CST IV (2.0× acquisition, 2.5× transmission) versus CST I. Baseline preterm birth risk (9%) was calibrated to South African data (39), with CST IV conferring 1.5–2-fold elevated PTB risk (11,14,28,29). Full model specification is in the Supplemental Appendix.

Interventions Standard-of-care metronidazole was delivered through: (1) symptomatic care (20% population coverage) (17,40); (2) prenatal symptomatic care (72% uptake) (39,41); and (3) asymptomatic prenatal screening (Nugent ≥7 treated) (41). LBP was introduced as adjunctive therapy following metronidazole. Three product characteristics were varied: efficacy of CST I restoration (50%, 65%, 80%) (23,24); duration of protection (6, 12, 18 months) (23–26); and microbiome stability (baseline: 90% reduction in reversion risk reflecting LBP strain fitness advantage; sensitivity: no fitness advantage) (3,25,26).

Outcomes and analysis Nine core scenarios (all efficacy-duration combinations) were evaluated. Number needed to treat (NNT) was calculated as LBP courses delivered divided by outcomes averted, contextualized against published benchmarks for oral PrEP (30,31) and vaginal progesterone (32). Sensitivity analyses using a reference profile (80% efficacy, 12-month duration) examined: microbiome stability, efficacy in refractory dysbiosis, introduction timing, expanded prenatal screening, direct LBP for intermediate dysbiosis (Nugent 4–6 without MTZ), improved metronidazole (82%→92%), and symptomatic care coverage (10%, 20%, 30%). We additionally examined the relationship between durable cure (Nugent 0–3 at 6 months) and modelled CST I prevalence across product profiles. Results are medians with 5th–95th percentile ranges across 50 stochastic replicates.

RESULTS

Model calibration The model reproduced South African HIV prevalence trends from 1990–2020 (37,38) and generated 60% BV prevalence with 50% recurrence at 12 months post-

metronidazole (1,2,5,6). Baseline projections (2,743,854 HIV infections and 2,312,147 PTBs over 2035–2050) align with UNAIDS trajectory estimates (37).

Trial endpoints and population health impact: the Nugent–CST I relationship Figure 1 shows modelled durable cure (Nugent 0–3 at 6 months) alongside CST I prevalence at 6 months across all nine product profiles. Products with longer duration consistently achieve higher durable cure rates regardless of efficacy (23,24), and products with high CST I prevalence show reasonable alignment between Nugent endpoints and population impact. However, products where Nugent normalization occurs through *L. iners* (CST III) dominance rather than *L. crispatus* engraftment (3,25) show substantial divergence between trial endpoints and population health impact — divergence that is undetectable from Nugent scores alone and increases as CST I prevalence decreases. These results motivate population impact modelling to translate trial signals into CST I prevalence estimates and downstream health outcomes, and support including metagenomic CST classification alongside Nugent scoring in Phase III trials.

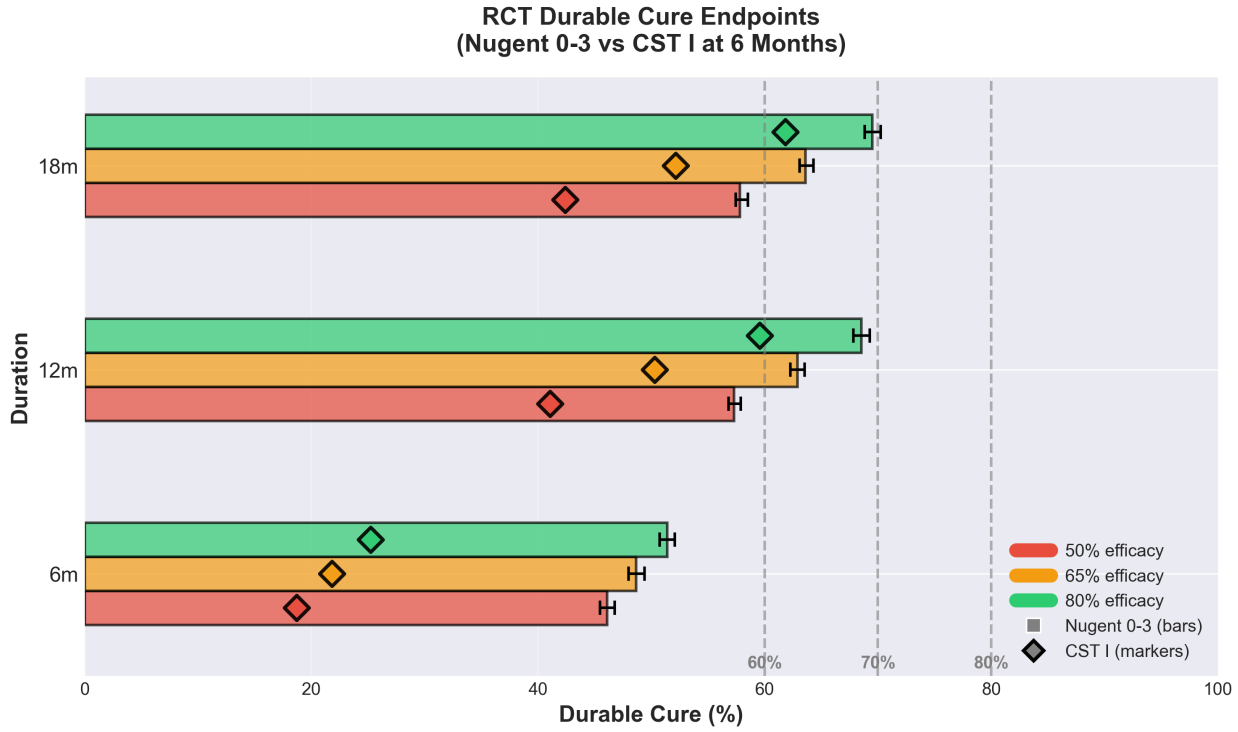

Figure 1: RCT durable cure endpoints (Nugent 0–3 vs. CST I prevalence at 6 months) across efficacy and duration combinations. Bar height shows Nugent 0–3 prevalence; diamond markers show CST I prevalence. Dashed vertical lines indicate candidate TPP thresholds at 60%, 70%, and 80% durable cure.

Primary impact estimates Under plausible product assumptions (80% efficacy, 12-month duration), LBP introduction in 2035 was projected to avert 173,000 HIV infections (6.5%, 95% UI: 5.0%–7.7%) and 251,000 preterm births (11.1%, 95% UI: 10.4%–11.8%) by 2050 (Table 1). NNT was 109 (95% UI: 89–150) for HIV prevention and 75 (95% UI: 68–78) for PTB prevention.

Table 1. Primary impact estimates (80% efficacy, 12-month duration)

Outcome	Baseline (2035–2050)	Averted	% Reduction	NNT
HIV infections	2,743,854	173,000 (127,000–209,000)	6.5% (5.0%–7.7%)	109 (89–150)
Preterm births	2,312,147	251,000 (241,000–274,000)	11.1% (10.4%–11.8%)	75 (68–78)

Impact across product characteristics HIV infections averted ranged from 42,000 (50% efficacy, 6-month duration; 1.5% reduction) to 249,000 (80% efficacy, 18-month duration; 9.0% reduction), and preterm births averted from 177,000 (7.6%) to 258,000 (11.2%) across all scenarios (Figure 2). HIV prevention showed a clear dose-response relationship with duration (4–6-fold variation at constant efficacy), whereas PTB prevention was largely insensitive to duration (≤5% variation), driven primarily by efficacy — a stark contrast with direct implications for TPP prioritization. For HIV prevention, meaningful impact (>5% reduction) required duration ≥12 months; 6-month scenarios yielded only 1.5–2.0% reductions regardless of efficacy. For PTB prevention, efficacy ≥50% achieved >7.5% reduction regardless of duration (11–14,29,30).

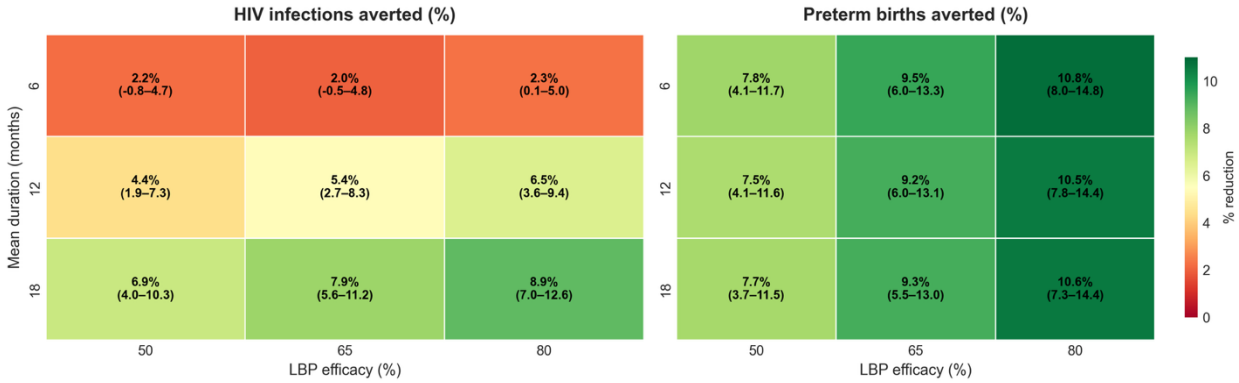

Figure 2: HIV infections averted (%) and preterm births averted (%) from 2035–2050, shown as paired panels across efficacy and duration combinations.

Number needed to treat HIV NNT ranged from 70 (80% efficacy, 18-month duration) to 498 (50% efficacy, 6-month duration), a 7-fold difference. PTB NNT ranged from 69 to 112, a 1.6-fold difference (Figure 3). At the reference scenario, the composite NNT to prevent one HIV infection or one PTB was 42. Extending duration from 6 to 18 months at 80% efficacy reduced HIV NNT 5.2-fold (362 to 70) but PTB NNT only 1.1-fold (79 to 69).

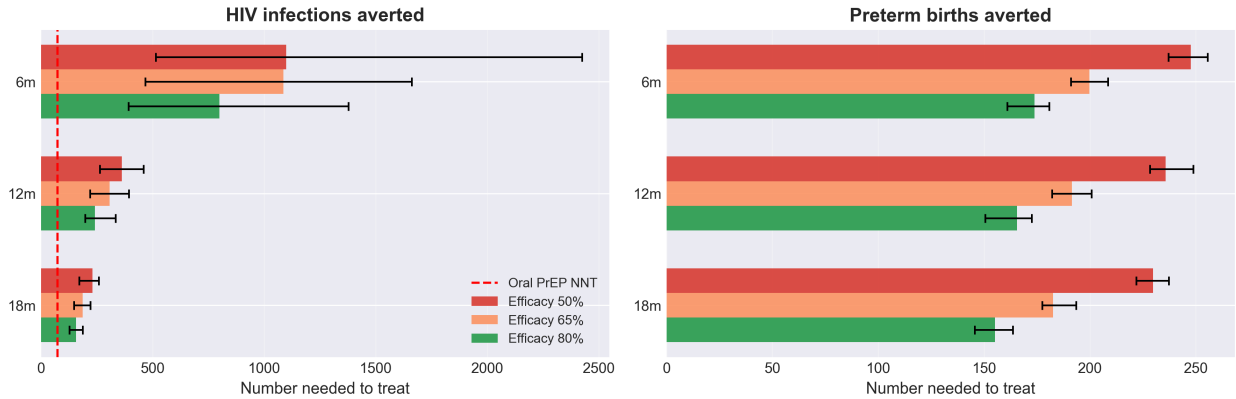

Figure 3: NNT across efficacy-duration combinations, paired panels for HIV (with oral PrEP benchmark, refs 31,32) and PTB.

Sensitivity analyses Microbiome stability was the largest single source of uncertainty: removing the LBP strain fitness advantage (reversion at endogenous rates) (25,26,38,39) reduced HIV infections averted from 173,000 to 84,000 (−51%) and PTBs averted from 251,000 to 162,000 (−36%), underscoring the importance of whether LBP induces durable ecological restructuring or requires retreatment. Achieving full LBP response in metronidazole non-responders (5,18) increased HIV infections averted to 200,000 (+16%) and PTBs averted to 305,000 (+21%), identifying refractory dysbiosis as a high-value development target. Delaying introduction from 2035 to 2040 reduced impact by 61% for HIV and 60% for PTB — a larger penalty than the shortened evaluation window alone, reflecting slower epidemic-level benefit accumulation (40,41). Delivery strategy substantially modified impact: symptomatic care coverage variation from 10% to 30% (17,43) produced a nearly 2-fold difference in HIV infections averted (98,000 to 182,000), highlighting population reach as a determinant of impact comparable to product efficacy. Adding asymptomatic prenatal screening (44) increased PTBs averted by 15%; further extending LBP to Nugent 4–6 pregnant women without MTZ pre-treatment produced an additional 24,000 PTBs averted — the largest absolute PTB gain of any scenario — reflecting that a meaningful share of PTB burden occurs in women with intermediate rather than severe dysbiosis (30). Improved metronidazole efficacy (82%→92%) (5,18) showed complementarity rather than substitution with LBP, yielding greater combined impact than either intervention alone (Figure 4).

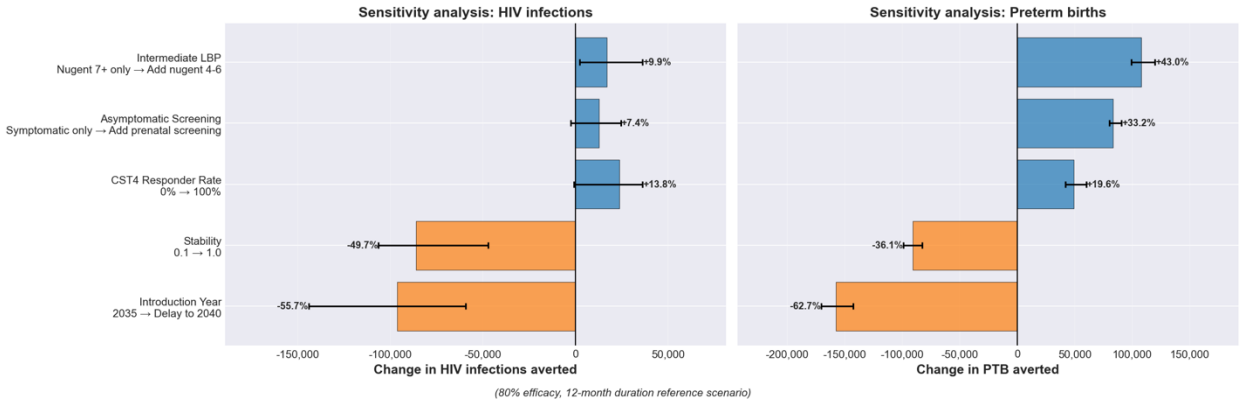

Figure 4: Tornado plot showing sensitivity of HIV infections averted (left panel) and preterm births averted (right panel), color-coded green (opportunities) and red/orange (risks).

Stratified impacts Impact concentrated among women aged 20–35 years (36,37): women aged 20–24 accounted for 32% of HIV infections averted; women aged 25–34 accounted for 48% of PTBs averted (Supplemental Figure S1). Among HIV-negative women, LBP prevented primary acquisition (87% of infections averted); among HIV-positive women, LBP reduced onward transmission (13% of infections averted).

DISCUSSION

This study quantified the population-level impact of LBPs for vaginal microbiome restoration and identified which product attributes and trial design choices are most predictive of population health benefit. Under plausible assumptions, LBP introduction in 2035 could avert 173,000 HIV infections and 251,000 preterm births by 2050, with impact determined as much by delivery reach and microbiome stability as by product efficacy.

Causal evidence and biological mechanisms The projected benefits depend on the assumption that sustained CST I dominance causally reduces HIV acquisition and PTB risk. This assumption is grounded in meta-analyses demonstrating 1.5–3-fold elevated HIV risk and 1.5–2-fold elevated PTB risk with vaginal dysbiosis (7,8,11,14); prospective cohort evidence linking genital inflammation, epithelial disruption, and reduced antimicrobial peptide activity to non-*Lactobacillus*-dominant states (9,10,28); mechanistic studies demonstrating that *L. crispatus* lactic acid production lowers vaginal pH and inhibits pathogen growth (21,22); and longitudinal evidence that CST I-to-CST IV transitions precede increased HIV susceptibility (25,26,28). For PTB, ascending infection and inflammatory pathways provide biologically plausible mechanisms (11–14,29,30). Residual confounding cannot be excluded, and if true causal effects are smaller than current estimates, population impact would proportionally decline. Future randomized trials with HIV and PTB endpoints will be critical for refining these projections.

The differential sensitivity of HIV versus PTB outcomes to duration reflects fundamental temporal differences. HIV prevention requires protection across years of sexual exposure, explaining the 4–6-fold variation in HIV impact across duration levels — a counterintuitive finding meaning a 50% efficacy/18-month product outperforms an 80% efficacy/6-month product for HIV prevention. PTB prevention is largely duration-insensitive because the relevant

risk window concentrates within the 9-month gestational period, making efficacy the dominant driver (11–14,29,30).

From trial endpoints to TPP targets Our finding that Nugent 0–3 is an imperfect proxy for CST I — with divergence increasing for products achieving normalization through *L. iners* rather than *L. crispatus* dominance (3,25) — has direct investment implications. Funding decisions based on Nugent endpoints alone risk systematic misallocation toward products with strong trial signals but modest population benefit. Metagenomic CST classification alongside Nugent scoring in Phase III trials is therefore not merely a scientific refinement but a necessary condition for reliable go/no-go decisions. Our threshold analyses provide exploratory quantitative TPP benchmarks: duration ≥12 months is a necessary condition for meaningful HIV prevention impact (>5% reduction); for PTB prevention, efficacy ≥50% achieves >7.5% reduction regardless of duration. Microbiome stability — a parameter not currently measured in trials — represents the largest source of uncertainty, with 35–51% impact variation between optimistic and pessimistic stability assumptions. Serial microbiome sampling at 12–18 months post-treatment should be a Phase III priority (see Supplemental Appendix for full trial design recommendations).

Delivery strategy as a determinant of impact Population impact is bounded as much by delivery reach as by product characteristics. The 2-fold difference in HIV infections averted across symptomatic care coverage scenarios (10–30%) demonstrates that expanding care-seeking through community outreach or task-shifting could amplify impact of a moderately effective product. Asymptomatic prenatal screening and extending LBP eligibility to intermediate dysbiosis represent the highest-leverage delivery investments for PTB prevention, together yielding a 24.7% total PTB gain over the base case — achievable without changes to the product itself.

Comparison with existing interventions The population-level HIV NNT of 109 under plausible assumptions is narrower than comparison with targeted interventions would suggest, because LBP is delivered through routine care rather than high-risk programs. Oral PrEP demonstrates NNTs of 13–40 in high-risk populations (31,32), but these are not directly comparable given different delivery models and populations. The composite NNT of 42 to prevent one HIV infection or one PTB better reflects the dual-benefit value proposition. For PTB prevention, vaginal progesterone demonstrates NNT of 11 in high-risk short-cervix populations (33), compared to our population-level NNT of 75 — reflecting delivery without high-risk screening. LBP addresses a biological risk factor not targeted by existing interventions and shows complementarity rather than competition with improved antibiotic regimens and PrEP (31,32).

Limitations Efficacy estimates (50–80%) reflect early-phase trial data (23,24); the stability parameter has 35–51% impact variation and requires empirical validation (3,25,26); we assumed high uptake and adherence without modelling access barriers (17,28); and the model simplifies microbiome complexity into three CSTs (3,25) with uncertain generalizability beyond South Africa (34,40,41). Projections through 2050 carry substantial uncertainty in epidemic trajectories and health system evolution (40,41). Results should inform development priorities and trial design but cannot substitute for clinical evidence.

CONCLUSIONS

LBPs could yield meaningful reductions in HIV infections (5–9%) and preterm births (8–11%) in high-burden settings (7–14,40), contingent on sustained colonization (≥12 months) and adequate efficacy (≥65%). Duration is the non-negotiable TPP attribute for HIV prevention; PTB prevention is achievable across a broader product profile range. The NNT under plausible assumptions (109 for HIV; 75 for PTB) compares favorably with population-level benchmarks (31–33), and the composite NNT of 42 reflects substantial dual-benefit value. Microbiome stability and delivery reach are as important as product characteristics. A critical methodological finding is that Nugent endpoints are imperfect proxies for CST I establishment, and Phase III trials should include metagenomic CST classification to enable reliable translation of trial signals into population health projections and investment decisions (3,25,26). Integration into existing BV treatment and prenatal care pathways is feasible (17,43,44), with real-world impact highly sensitive to the proportion of women reached through symptomatic and prenatal channels.

AUTHOR CONTRIBUTIONS

JAC, LRD, and SLA conceptualized the study. JAC and LRD developed the model. JAC conducted the analyses. JAC, LRD, HS, CS, LS, and SLA interpreted results. JAC drafted the manuscript. All authors critically reviewed and approved the final manuscript.

DECLARATION OF INTERESTS

All authors are employed by the Gates Foundation. The authors declare no other competing interests.

DATA AVAILABILITY

Model code will be made publicly available at https://github.com/starsimhub/hiv_vmb_southafrica upon publication.

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